



Variables on Both Sides Teaching Bundle

Grade 8 • North Carolina • Mathematics

NC.8.EE.7 • Solve and classify linear equations.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Solve and classify linear equations. This bundle practices a focused part of NC.8.EE.7; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Collect variable terms on one side and constants on the other. An always-true statement has infinitely many solutions; a contradiction has none.

Example 1: Solve $12x + 4 = 7x + (-1)$. Show your work and check.

Subtract $7x$ and 4 : $5x = -5$. Divide by 5 : $x = -1$. Both original sides equal -8 .

Example 2: Solve $11x + 5 = 8x + (35)$. Show your work and check.

Subtract $8x$ and 5 : $3x = 30$. Divide by 3 : $x = 10$. Both original sides equal 115 .

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1-4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

Solve $7x + 6 = 7x + 6$. Classify the solution set and justify it.

Explain your method.

CARD 02

Solve $6x + 2 = 6x + 5$. Classify the solution set and justify it.

Explain your method.

CARD 03

Solve $14x + 8 = 9x + (38)$. Show your work and check.

Explain your method.

CARD 04

Solve $15x + 3 = 8x + (-18)$. Show your work and check.

Explain your method.



Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

Solve $8x + 3 = 8x + 3$. Classify the solution set and justify it.

Explain your method.

CARD 06

Solve $4x + 3 = 4x + 7$. Classify the solution set and justify it.

Explain your method.

CARD 07

Solve $7x + 8 = 2x + (-7)$. Show your work and check.

Explain your method.

CARD 08

Solve $15x + 3 = 8x + (-18)$. Show your work and check.

Explain your method.



Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

Solve $9x + 6 = 9x + 6$. Classify the solution set and justify it.

Explain your method.

CARD 10

Solve $6x + 6 = 6x + 9$. Classify the solution set and justify it.

Explain your method.

CARD 11

Solve $10x + 3 = 7x + (3)$. Show your work and check.

Explain your method.

CARD 12

Solve $10x + 3 = 8x + (5)$. Show your work and check.

Explain your method.



Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

Solve $9x + 6 = 9x + 6$. Classify the solution set and justify it.

Explain your method.

CARD 14

Solve $8x + 3 = 8x + 7$. Classify the solution set and justify it.

Explain your method.

CARD 15

Solve $8x + 5 = 2x + (-7)$. Show your work and check.

Explain your method.

CARD 16

Solve $10x + 4 = 6x + (-8)$. Show your work and check.

Explain your method.



Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. Solve $12x + 5 = 8x + (25)$. Show your work and check.

2. Solve $12x + 2 = 5x + (58)$. Show your work and check.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. Solve $5x + 4 = 5x + 4$. Classify the solution set and justify it.

4. Solve $3x + 4 = 3x + 9$. Classify the solution set and justify it.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. Solve $7x + 6 = 3x + (26)$. Show your work and check.

6. Solve $10x + 5 = 7x + (29)$. Show your work and check.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. Solve $9x + 4 = 9x + 4$. Classify the solution set and justify it.

8. Solve $9x + 8 = 9x + 14$. Classify the solution set and justify it.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. Solve $9x + 2 = 2x + (-5)$. Show your work and check.

10. Solve $16x + 3 = 9x + (24)$. Show your work and check.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. Solve $13x + 8 = 8x + (43)$. Show your work and check.

2. Solve $7x + 3 = 2x + (53)$. Show your work and check.

3. Solve $6x + 2 = 6x + 2$. Classify the solution set and justify it.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. Solve $7x + 4 = 7x + 9$. Classify the solution set and justify it.

5. Solve $9x + 6 = 4x + (6)$. Show your work and check.

6. Solve $10x + 5 = 5x + (-10)$. Show your work and check.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. Solve $5x + 7 = 3x + (17)$. Show your work and check.

2. Solve $10x + 8 = 3x + (-6)$. Show your work and check.

3. Solve $6x + 3 = 6x + 3$. Classify the solution set and justify it.

4. Solve $6x + 5 = 6x + 11$. Classify the solution set and justify it.



Task-card worked solutions

1. All real numbers

Subtract $7x$ from each side to obtain $6 = 6$, true for every real x .

2. No solution

Subtract $6x$ to obtain $2 = 5$, a contradiction.

3. $x = 6$

Subtract $9x$ and 8 : $5x = 30$. Divide by 5 : $x = 6$. Both original sides equal 92 .

4. $x = -3$

Subtract $8x$ and 3 : $7x = -21$. Divide by 7 : $x = -3$. Both original sides equal -42 .

5. All real numbers

Subtract $8x$ from each side to obtain $3 = 3$, true for every real x .

6. No solution

Subtract $4x$ to obtain $3 = 7$, a contradiction.

Task-card worked solutions

7. $x = -3$

Subtract $2x$ and 8 : $5x = -15$. Divide by 5 : $x = -3$. Both original sides equal -13 .

8. $x = -3$

Subtract $8x$ and 3 : $7x = -21$. Divide by 7 : $x = -3$. Both original sides equal -42 .

9. All real numbers

Subtract $9x$ from each side to obtain $6 = 6$, true for every real x .

10. No solution

Subtract $6x$ to obtain $6 = 9$, a contradiction.

11. $x = 0$

Subtract $7x$ and 3 : $3x = 0$. Divide by 3 : $x = 0$. Both original sides equal 3 .

12. $x = 1$

Subtract $8x$ and 3 : $2x = 2$. Divide by 2 : $x = 1$. Both original sides equal 13 .

Task-card worked solutions

13. All real numbers

Subtract $9x$ from each side to obtain $6 = 6$, true for every real x .

14. No solution

Subtract $8x$ to obtain $3 = 7$, a contradiction.

15. $x = -2$

Subtract $2x$ and 5 : $6x = -12$. Divide by 6 : $x = -2$. Both original sides equal -11 .

16. $x = -3$

Subtract $6x$ and 4 : $4x = -12$. Divide by 4 : $x = -3$. Both original sides equal -26 .

Warm-up worked solutions

1. $x = 5$

Subtract $8x$ and 5 : $4x = 20$. Divide by 4 : $x = 5$. Both original sides equal 65 .

2. $x = 8$

Subtract $5x$ and 2 : $7x = 56$. Divide by 7 : $x = 8$. Both original sides equal 98 .

3. All real numbers

Subtract $5x$ from each side to obtain $4 = 4$, true for every real x .

4. No solution

Subtract $3x$ to obtain $4 = 9$, a contradiction.

5. $x = 5$

Subtract $3x$ and 6 : $4x = 20$. Divide by 4 : $x = 5$. Both original sides equal 41 .

6. $x = 8$

Subtract $7x$ and 5 : $3x = 24$. Divide by 3 : $x = 8$. Both original sides equal 85 .

Warm-up worked solutions

7. All real numbers

Subtract $9x$ from each side to obtain $4 = 4$, true for every real x .

8. No solution

Subtract $9x$ to obtain $8 = 14$, a contradiction.

9. $x = -1$

Subtract $2x$ and 2 : $7x = -7$. Divide by 7 : $x = -1$. Both original sides equal -7 .

10. $x = 3$

Subtract $9x$ and 3 : $7x = 21$. Divide by 7 : $x = 3$. Both original sides equal 51 .



Classwork worked solutions

1. $x = 7$

Subtract $8x$ and 8 : $5x = 35$. Divide by 5 : $x = 7$. Both original sides equal 99 .

2. $x = 10$

Subtract $2x$ and 3 : $5x = 50$. Divide by 5 : $x = 10$. Both original sides equal 73 .

3. All real numbers

Subtract $6x$ from each side to obtain $2 = 2$, true for every real x .

4. No solution

Subtract $7x$ to obtain $4 = 9$, a contradiction.

5. $x = 0$

Subtract $4x$ and 6 : $5x = 0$. Divide by 5 : $x = 0$. Both original sides equal 6 .

6. $x = -3$

Subtract $5x$ and 5 : $5x = -15$. Divide by 5 : $x = -3$. Both original sides equal -25 .

Exit-ticket worked solutions

1. $x = 5$

Subtract $3x$ and 7 : $2x = 10$. Divide by 2 : $x = 5$. Both original sides equal 32 .

2. $x = -2$

Subtract $3x$ and 8 : $7x = -14$. Divide by 7 : $x = -2$. Both original sides equal -12 .

3. All real numbers

Subtract $6x$ from each side to obtain $3 = 3$, true for every real x .

4. No solution

Subtract $6x$ to obtain $5 = 11$, a contradiction.

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. Solve $7x + 2 = 7x + 2$. Classify the solution set and justify it.

2. Solve $8x + 4 = 8x + 8$. Classify the solution set and justify it.

3. Solve $10x + 2 = 3x + (58)$. Show your work and check.

4. Solve $11x + 4 = 5x + (52)$. Show your work and check.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. Solve $3x + 8 = 3x + 8$. Classify the solution set and justify it.

6. Solve $7x + 2 = 7x + 8$. Classify the solution set and justify it.

7. Solve $6x + 6 = 4x + (0)$. Show your work and check.

8. Solve $12x + 8 = 7x + (18)$. Show your work and check.



Fall • Worked solutions

1. All real numbers

Subtract $7x$ from each side to obtain $2 = 2$, true for every real x .

2. No solution

Subtract $8x$ to obtain $4 = 8$, a contradiction.

3. $x = 8$

Subtract $3x$ and 2 : $7x = 56$. Divide by 7 : $x = 8$. Both original sides equal 82 .

4. $x = 8$

Subtract $5x$ and 4 : $6x = 48$. Divide by 6 : $x = 8$. Both original sides equal 92 .

Fall • Worked solutions

5. All real numbers

Subtract $3x$ from each side to obtain $8 = 8$, true for every real x .

6. No solution

Subtract $7x$ to obtain $2 = 8$, a contradiction.

7. $x = -3$

Subtract $4x$ and 6 : $2x = -6$. Divide by 2 : $x = -3$. Both original sides equal -12 .

8. $x = 2$

Subtract $7x$ and 8 : $5x = 10$. Divide by 5 : $x = 2$. Both original sides equal 32 .



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. Solve $4x + 7 = 2x + (3)$. Show your work and check.

2. Solve $14x + 2 = 9x + (37)$. Show your work and check.

3. Solve $4x + 7 = 4x + 7$. Classify the solution set and justify it.

4. Solve $7x + 7 = 7x + 12$. Classify the solution set and justify it.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. Solve $12x + 3 = 7x + (-17)$. Show your work and check.

6. Solve $11x + 7 = 6x + (47)$. Show your work and check.

7. Solve $2x + 7 = 2x + 7$. Classify the solution set and justify it.

8. Solve $3x + 5 = 3x + 11$. Classify the solution set and justify it.



Winter • Worked solutions

1. $x = -2$

Subtract $2x$ and 7 : $2x = -4$. Divide by 2 : $x = -2$. Both original sides equal -1 .

2. $x = 7$

Subtract $9x$ and 2 : $5x = 35$. Divide by 5 : $x = 7$. Both original sides equal 100 .

3. All real numbers

Subtract $4x$ from each side to obtain $7 = 7$, true for every real x .

4. No solution

Subtract $7x$ to obtain $7 = 12$, a contradiction.

Winter • Worked solutions

5. $x = -4$

Subtract $7x$ and 3 : $5x = -20$. Divide by 5 : $x = -4$. Both original sides equal -45 .

6. $x = 8$

Subtract $6x$ and 7 : $5x = 40$. Divide by 5 : $x = 8$. Both original sides equal 95 .

7. All real numbers

Subtract $2x$ from each side to obtain $7 = 7$, true for every real x .

8. No solution

Subtract $3x$ to obtain $5 = 11$, a contradiction.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. Solve $5x + 7 = 5x + 7$. Classify the solution set and justify it.

2. Solve $4x + 3 = 4x + 7$. Classify the solution set and justify it.

3. Solve $13x + 3 = 6x + (-25)$. Show your work and check.

4. Solve $14x + 8 = 7x + (1)$. Show your work and check.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. Solve $6x + 5 = 6x + 5$. Classify the solution set and justify it.

6. Solve $9x + 6 = 9x + 12$. Classify the solution set and justify it.

7. Solve $9x + 4 = 7x + (20)$. Show your work and check.

8. Solve $11x + 5 = 8x + (-10)$. Show your work and check.



Spring break • Worked solutions

1. All real numbers

Subtract $5x$ from each side to obtain $7 = 7$, true for every real x .

2. No solution

Subtract $4x$ to obtain $3 = 7$, a contradiction.

3. $x = -4$

Subtract $6x$ and 3 : $7x = -28$. Divide by 7 : $x = -4$. Both original sides equal -49 .

4. $x = -1$

Subtract $7x$ and 8 : $7x = -7$. Divide by 7 : $x = -1$. Both original sides equal -6 .



Spring break • Worked solutions

5. All real numbers

Subtract $6x$ from each side to obtain $5 = 5$, true for every real x .

6. No solution

Subtract $9x$ to obtain $6 = 12$, a contradiction.

7. $x = 8$

Subtract $7x$ and 4 : $2x = 16$. Divide by 2 : $x = 8$. Both original sides equal 76 .

8. $x = -5$

Subtract $8x$ and 5 : $3x = -15$. Divide by 3 : $x = -5$. Both original sides equal -50 .



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. Solve $6x + 2 = 2x + (-2)$. Show your work and check.

2. Solve $6x + 2 = 3x + (29)$. Show your work and check.

3. Solve $3x + 6 = 3x + 6$. Classify the solution set and justify it.

4. Solve $5x + 5 = 5x + 11$. Classify the solution set and justify it.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. Solve $9x + 5 = 6x + (8)$. Show your work and check.

6. Solve $14x + 5 = 9x + (20)$. Show your work and check.

7. Solve $8x + 3 = 8x + 3$. Classify the solution set and justify it.

8. Solve $8x + 7 = 8x + 10$. Classify the solution set and justify it.



Summer • Worked solutions

1. $x = -1$

Subtract $2x$ and 2 : $4x = -4$. Divide by 4 : $x = -1$. Both original sides equal -4 .

2. $x = 9$

Subtract $3x$ and 2 : $3x = 27$. Divide by 3 : $x = 9$. Both original sides equal 56 .

3. All real numbers

Subtract $3x$ from each side to obtain $6 = 6$, true for every real x .

4. No solution

Subtract $5x$ to obtain $5 = 11$, a contradiction.



Summer • Worked solutions

5. $x = 1$

Subtract $6x$ and 5 : $3x = 3$. Divide by 3 : $x = 1$. Both original sides equal 14 .

6. $x = 3$

Subtract $9x$ and 5 : $5x = 15$. Divide by 5 : $x = 3$. Both original sides equal 47 .

7. All real numbers

Subtract $8x$ from each side to obtain $3 = 3$, true for every real x .

8. No solution

Subtract $8x$ to obtain $7 = 10$, a contradiction.



Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Solve and classify linear equations. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Collect variable terms on one side and constants on the other. An always-true statement has infinitely many solutions; a contradiction has none.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. Solve $5x + 4 = 5x + 4$. Classify the solution set and justify it.

2. Solve $5x + 6 = 5x + 10$. Classify the solution set and justify it.

3. Solve $8x + 7 = 5x + (4)$. Show your work and check.

4. Solve $8x + 7 = 6x + (11)$. Show your work and check.



Substitute lesson • Worked solutions

1. All real numbers

Subtract $5x$ from each side to obtain $4 = 4$, true for every real x .

2. No solution

Subtract $5x$ to obtain $6 = 10$, a contradiction.

3. $x = -1$

Subtract $5x$ and 7 : $3x = -3$. Divide by 3 : $x = -1$. Both original sides equal -1 .

4. $x = 2$

Subtract $6x$ and 7 : $2x = 4$. Divide by 2 : $x = 2$. Both original sides equal 23 .

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Solve and classify linear equations. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Collect variable terms on one side and constants on the other. An always-true statement has infinitely many solutions; a contradiction has none.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. Solve $5x + 3 = 5x + 3$. Classify the solution set and justify it.

2. Solve $7x + 3 = 7x + 10$. Classify the solution set and justify it.

3. Solve $12x + 2 = 8x + (14)$. Show your work and check.

4. Solve $11x + 4 = 9x + (0)$. Show your work and check.



Emergency substitute lesson • Worked solutions

1. All real numbers

Subtract $5x$ from each side to obtain $3 = 3$, true for every real x .

2. No solution

Subtract $7x$ to obtain $3 = 10$, a contradiction.

3. $x = 3$

Subtract $8x$ and 2 : $4x = 12$. Divide by 4 : $x = 3$. Both original sides equal 38 .

4. $x = -2$

Subtract $9x$ and 4 : $2x = -4$. Divide by 2 : $x = -2$. Both original sides equal -18 .

A completed lesson-plan example

Level and focus

Grade 8; NC.8.EE.7; Solve and classify linear equations.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.8.EE.7 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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